Cristhian López Zambrano

AI Engineer · Agentic Systems & LLM Applications

Guayaquil, Ecuador · (+593) 990-901765 · thianlopezz@gmail.com · linkedin.com/in/thianlopezz · thianlopezz.com

SUMMARY

AI engineer focused on agentic systems and LLM applications, with software engineering experience since 2014. At Parallel Public Works I run an agentic development pipeline: AI coding agents take tasks from pickup to pull request, with human review before any merge. I have also set up MCP servers and implemented agent skills, and I work hands-on with Claude Code, Codex and ChatGPT. Independently, I built Astillero, a Python multi-agent orchestration framework, and run a local LLM lab with Ollama and RAG on pgvector.

EXPERIENCE

Parallel Public Works (initially via Jobsity) · Full Stack Developer

Jan 2022 – Present

- Run an agentic development pipeline on the company codebase: Claude Code and Codex agents, working in parallel through Orca and herdr, take tasks from pickup to pull request, with human-in-the-loop review required before any merge.
- Set up MCP servers and implemented agent skills for development tasks.
- Revamped the product architecture to speed up Gatsby static file generation for large web pages (Node.js), cutting build time by 60%.
- Cut development environment setup time by 80% by introducing DevContainers and Docker.
- Integrated Auth0 to link a decoupled Gatsby frontend with a Drupal backend, and migrated legacy data from 4 projects into new Drupal content sites using built-in and custom plugins.

INDEPENDENT AI PROJECTS

Astillero · Multi-agent orchestration framework (Python, private repo)

- Built a Kanban-based framework that coordinates AI agents through a deterministic orchestrator, organized in three layers: board state, agent role contracts and the orchestrator loop.
- Uses only the Python standard library; a CLI handles Product Owner interactions, and key policies are verified with mock agents.

Local LLM lab

- Built a home lab that runs local LLMs on Ollama, with RAG over my own documents (PostgreSQL, pgvector) and agent-based automations.

LLM statement reconciliation

- Built a monthly workflow where an LLM classifies transactions from credit card statement PDFs (three issuers) as confirmed, uncertain or excluded, and outputs a formatted Excel reconciliation for a shared expense fund.
- Added a merchant memory workbook that keeps classifications consistent across months; unresolved conflicts are flagged for review.

EARLIER EXPERIENCE

Galápagos Government Council · Systems Information Analyst

Jul 2021 – Dec 2021

- Built a web application (React, Laravel, Node.js, Docker) for the institution's digital transformation program, digitizing historical and manual procedures.

Liris S.A. · Mobile Application Analyst

May 2021 – Jul 2021

- Maintained and extended two Ionic 3 apps: an internal employee app and an e-commerce app with more than 14,000 users.

JSS Technologies · Full Stack Developer

Jan 2021 – May 2021

- Built a fintech sales web app (React, Laravel) with Twilio video calls and end-to-end automation of prospecting, contract generation, digital signing and reporting.

Bitbox · Systems Integrator

May 2020 – Jan 2021

- Built a self-checkout app for an insurance company (Ionic, WooCommerce API) that automated policy sales and removed manual work from the vehicle inspection process.
- Built a Laravel and Tailwind CSS microsite for 50,000+ customers to check eligibility for digital coupons.

Ecudevs · Founder, Instructor and Developer

Sep 2018 – Apr 2020

- Built a Laravel system to automate educational program planning and tracking.
- Designed 3 learning programs and taught JavaScript, Angular, React and Ionic courses.
- Product Owner of Nomadem, a web and mobile app connecting travelers with local agencies in the Galápagos Islands.

Sipecom · Semi-Senior .NET Programmer

Jan 2016 – Jun 2017

- Built a C# process that generates and emails account statements for 500,000+ customers of one of Ecuador's largest banks, tuning it to cut processing time.
- Shipped an employee travel-expense app (ASP.NET MVC, jQuery) that cut the time to start expense reimbursements by 30%.

Fondo de Cesantía del Magisterio Ecuatoriano · Junior .NET Programmer

Oct 2014 – Mar 2015

- Developed and maintained web applications with C# and ASP.NET.

SKILLS

AI coding tools: Claude Code, OpenAI Codex, ChatGPT, Orca and herdr (for running coding agents in parallel)
AI and agents: Agentic development workflows, multi-agent orchestration, Claude Agent SDK, Model Context Protocol (MCP) servers, agent skills, LLM APIs (Anthropic Claude, OpenAI), human-in-the-loop review
RAG and local models: Retrieval-augmented generation (RAG), vector databases (PostgreSQL with pgvector), Ollama, local LLMs
Programming languages: Python, JavaScript, PHP, C#
Frameworks and tools: Node.js, React, Gatsby, Angular, Laravel, Drupal, React Native (Expo), Docker, DevContainers, Nginx
Data and cloud: PostgreSQL, MySQL, SQL Server, AWS, Google Cloud Platform, Firebase, DigitalOcean
Languages: English (advanced), Spanish (native)

EDUCATION

Universidad Católica de Santiago de Guayaquil · Bachelor of Science in Systems Engineering · Oct 2018